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Inductor component

Abstract

An inductor component includes an element body. The element body has a mounting surface and a top surface that are parallel to each other, first and second parallel end surfaces, and first and second parallel side surfaces. An inductor wiring is disposed inside the element body. A first lead-out electrode is connected to a first end of the inductor wiring in an extension direction of the inductor wiring. A second lead-out electrode is connected to a second end of the inductor wiring in the extension direction of the inductor wiring. A lower surface of the first lead-out electrode and a lower surface of the second lead-out electrode are exposed from the mounting surface of the element body. The first lead-out electrode has a substantially quadrangular columnar shape that extends in a height direction. The first lead-out electrode is exposed from the first end surface and the first side surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Japanese Patent Application No. 2020-174615 filed Oct. 16, 2020, the entire content of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to an inductor component.

Background Art

(3) An inductor component disclosed in Japanese Unexamined Patent Application Publication No. 2019-057580 includes an element body. An inductor wiring is disposed inside the element body. A first lead-out electrode is connected to a first end of the inductor wiring. The first lead-out electrode extends in an L-shape across a mounting surface of the element body and a first end surface of the element body that is connected to the mounting surface. The surface of the first lead-out electrode is exposed from the mounting surface and the first end surface of the element body. A second lead-out electrode is connected to a second end of the inductor wiring. Similarly to the first lead-out electrode, the second lead-out electrode also extends in an L-shape across the mounting surface and a second end surface connected to the mounting surface. The surface of the second lead-out electrode is exposed from the mounting surface and the second end surface of the element body.

SUMMARY

(4) When lead-out electrodes extend along the mounting surface and end surfaces of the element body as in the inductor component disclosed in Japanese Unexamined Patent Application Publication No. 2019-057580, the lead-out electrodes cover large areas around the inductor wiring. Therefore, there is a risk that the lead-out electrodes will excessively block magnetic flux generated when a current flows along the inductor wiring.

(5) Accordingly, an aspect of the present disclosure provides an inductor component that includes an element body having a mounting surface and a top surface that are parallel to each other, a first end surface and a second end surface that are parallel to each other, and a first side surface and a second side surface that are parallel to each other; an inductor wiring disposed inside the element body; a first lead-out electrode connected to a first end of the inductor wiring; and a second lead-out electrode connected to a second end of the inductor wiring. Part of the first lead-out electrode and part of the second lead-out electrode are exposed from the mounting surface. The first lead-out electrode has a columnar shape that extends in a direction perpendicular to the mounting surface, and is exposed from the first end surface and the first side surface.

(6) According to this configuration, the first lead-out electrode has a columnar shape and is exposed from the first end surface and the first side surface. In other words, the first lead-out electrode is disposed in the form of a column at a position shifted towards a ridge line between the first end surface and the first side surface. Therefore, the extent to which the inductor wiring disposed inside the element body is covered by the first lead-out electrode can be reduced compared with a case where the first lead-out electrode has an L shape that extends onto the mounting surface and covers a large area of the first end surface. As a result, a situation in which magnetic flux generated when a current flows along the inductor wiring is excessively blocked by the first lead-out electrode is suppressed.

(7) Blocking of magnetic flux by the first lead-out electrode of the inductor component is suppressed.

(8) Other features, elements, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of preferred embodiments of the present disclosure with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an inductor component;
- (2) FIG. 2 is a side view of the inductor component;
- (3) FIG. 3 is a bottom view of the inductor component;
- (4) FIG. 4 is a top view of the internal structure of the inductor component;
- (5) FIG. 5 is a side view of the internal structure of the inductor component;
- (6) FIG. 6 is a sectional view along line 6-6 in FIG. 4;

(7) FIG. 7 is a sectional view along line 7-7 in FIG. 4; and

(8) FIG. 8 is a side view of the internal structure of an inductor component of a modification.

DETAILED DESCRIPTION

(9) Hereafter, an inductor component according to an embodiment will be described. In the drawings, constituent elements may be illustrated in an enlarged manner for ease of understanding. The dimensional ratios of the constituent elements may differ from the actual ratios or may differ from the ratios in other drawings.

(10) As illustrated in FIG. 1, an inductor component **10** includes an element body **20**. The element body **20** has a substantially rectangular parallelepiped shape on the whole. The element body **20** is composed of an insulator such as glass, resin, or alumina. As illustrated in FIG. 2, among the outer surfaces of the element body **20**, one surface is a mounting surface **21** and the surface that faces the mounting surface **21** is a top surface **22**. Therefore, the mounting surface **21** and the top surface **22** are parallel to each other. The mounting surface **21** is the surface that faces the circuit board when the inductor component **10** is mounted on a circuit board. The mounting surface **21** and the top surface **22** both have substantially rectangular shapes with substantially the same dimensions. “Substantially the same dimensions” means that the dimensions should be practically identical with errors of around 10 μm , for example, which may be due to manufacturing variations and so on, being acceptable.

(11) In the following description, a direction perpendicular to the mounting surface **21** is taken to be a height direction T_d , a side where the top surface **22** is disposed in the height direction T_d is referred to as an upper side, and a side where the mounting surface **21** is disposed in the height direction T_d is referred to as a lower side. In addition, a longitudinal direction of the mounting surface **21** is taken to be a length direction L_d and a lateral direction of the mounting surface **21** is taken to be a width direction W_d . In addition, as illustrated in FIG. 2, among the outer surfaces of the element body **20** that extend in a direction perpendicular to the mounting surface **21**, the surface on the side near a first end in the length direction L_d is taken to be a first end surface **23** and the surface on the side near a second end in the length direction L_d is taken to be a second end surface **24**. Therefore, the first end surface **23** and the second end surface **24** are parallel to each other.

(12) Furthermore, as illustrated in FIG. 3, among the outer surfaces of the element body **20** that extend in a direction perpendicular to the mounting surface **21**, the surface on the side near a first end in the width direction W_d is taken to be a first side surface **25** and the surface on the side near a second end in the width direction W_d is taken to be a second side surface **26**. Therefore, the first side surface **25** and the second side surface **26** are parallel to each other. In this embodiment, the dimension of the element body **20** in the length direction L_d is around 400 μm . The dimension of the element body **20** in the width direction W_d is around 200 μm . The dimension of the element body **20** in the height direction T_d is around 200 μm .

(13) As illustrated in FIG. 4, the inductor component **10** includes an inductor wiring **30**, a first lead-out electrode **40**, and a second lead-out electrode **50**. The inductor wiring **30** is disposed inside the element body **20**. The inductor wiring **30** is composed of an electrically conductive material such as silver or copper. In FIGS. 4 and 5, the internal structure of the element body **20** is illustrated with the element body **20** being depicted in a see-through manner.

(14) As illustrated in FIG. 5, when looking in the width direction W_d , the inductor wiring **30** extends along a substantially quadrangular path TR . When looking in the width direction W_d , the inductor wiring **30** includes first straight parts **31**, which correspond to the side near the first end in the length direction L_d , second straight parts **32**, which correspond to the side near the second end in the length direction L_d , third straight parts **33**, which correspond to the side on the upper side in the height direction T_d , and fourth straight parts **34**, which correspond to the side on the lower side in the height direction T_d . Furthermore, as illustrated in FIG. 4, the inductor wiring **30** is formed of five conductive layers, namely, first to fifth conductor layers L_1 to L_5 , which are stacked in the width direction W_d , and vias **35** that connect the five first to fifth conductor layers L_1 to L_5 to

each other in the width direction Wd.

(15) As illustrated in FIG. 6, the first conductive layer L1, which is located on the side near the first end in the width direction Wd, includes the second straight part 32, the third straight part 33, and the fourth straight part 34. The second straight part 32, the third straight part 33, and the fourth straight part 34 of the first conductive layer L1 are disposed on a plane parallel to the first side surface 25. The third straight part 33 extends in the length direction Ld on a plane parallel to the first side surface 25. The third straight part 33 has a substantially quadrangular columnar shape. The third straight part 33 is located nearer the upper side in the height direction Td than the center of the element body 20.

(16) The second straight part 32 is connected to an end, which is on the side nearer the second end in the length direction Ld, of the third straight part 33 of the first conductive layer L1. The second straight part 32 has a substantially quadrangular columnar shape and extends in the height direction Td. The dimension of the second straight part 32 in the extension direction thereof is smaller than that of the third straight part 33. The second straight part 32 is located nearer the second end in the length direction Ld than the center of the element body 20.

(17) The fourth straight part 34 is connected to an end, which on the lower side in the height direction Td, of the second straight part 32 of the first conductive layer L1. The fourth straight part 34 has a substantially quadrangular columnar shape and extends in the length direction Ld. The dimension of the fourth straight part 34 of the first conductive layer L1 in the length direction Ld is smaller than the dimension of the third straight part 33 in the length direction Ld. Therefore, an end, which is on the side nearer the first end in the length direction Ld, of the fourth straight part 34 of the first conductive layer L1 is located nearer the second end in the length direction Ld than an end, which is on the side nearer the first end in the length direction Ld, of the third straight part 33 of the first conductive layer L1. As illustrated in FIG. 4, the via 35 extends from an end, which is on the side nearer the first end in the length direction Ld, of the fourth straight part 34 of the first conductive layer L1 toward the second end in the width direction Wd.

(18) The fourth straight part 34 of the second conductive layer L2 is connected to an end, which on the side nearer the second end in the width direction Wd, of the via 35 of the first conductive layer L1. The fourth straight part 34 is connected at first end portion 34a, which is a part thereof on the side nearer the first end in the length direction Ld.

(19) As illustrated in FIG. 7, the second conductive layer L2 includes the first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34. The first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34 of the second conductive layer L2 are disposed on a plane parallel to the first side surface 25. The first end portion 34a of the second conductive layer L2 extends in the length direction Ld so as to match the extension direction of the fourth straight part 34 of the first conductive layer L1 when looking in the width direction Wd.

(20) The first straight part 31 is connected to an end, which is on the side nearer the first end in the length direction Ld, of the first end portion 34a of the second conductive layer L2. The first straight part 31 of the second conductive layer L2 has a substantially quadrangular columnar shape and extends in the height direction Td.

(21) The third straight part 33 is connected to an end, which on the upper side in the height direction Td, of the first straight part 31 of the second conductive layer L2. The third straight part 33 of the second conductive layer L2 overlaps the third straight part 33 of the first conductive layer L1 when looking in the width direction Wd.

(22) The second straight part 32 is connected to an end, which nearer the second end in the length direction Ld, of the third straight part 33 of the second conductive layer L2. The second straight part 32 of the second conductive layer L2 overlaps the second straight part 32 of the first conductive layer L1 when looking in the width direction Wd.

(23) A second end portion 34b, which is a part of the fourth straight part 34 on the side nearer the

second end in the length direction Ld, is connected to an end, which is on the lower side in the height direction Td, of the second straight part 32 of the second conductive layer L2. The second end portion 34b has a substantially quadrangular columnar shape and extends in the length direction Ld. The end of the second end portion 34b on the side nearer the first end in the length direction Ld does not reach the first end portion 34a and the second end portion 34b and the first end portion 34a are separated from each other. When looking in the width direction Wd, the first end portion 34a of the fourth straight part 34 of the second conductive layer L2 lies within the region occupied by the fourth straight part 34 of the first conductive layer L1.

(24) As illustrated in FIG. 4, the via 35 extends from an end, which is on the side nearer the first end in the length direction Ld, of the second end portion 34b of the second conductive layer L2 toward the second end in the width direction Wd. The first end portion 34a, which is the part on the side near the first end in the length direction Ld, of the fourth straight part 34 of the third conductive layer L3 of the inductor wiring 30 is connected to this via 35. The third conductive layer L3 includes the first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34. The first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34 of the third conductive layer L3 are disposed on a plane parallel to the first side surface 25. Similarly to the second conductive layer L2, the third conductive layer L3 extends so as to draw a substantially quadrangular path TR formed of the first end portion 34a, the first straight part 31, the third straight part 33, the second straight part 32, and the second end portion 34b. The position of the gap between the first end portion 34a and the second end portion 34b in the third conductive layer L3 is nearer the second end in the length direction Ld than the position of the gap between the first end portion 34a and the second end portion 34b in the second conductive layer L2.

(25) The via 35 extends from an end, which is on the side nearer the first end in the length direction Ld, of the second end portion 34b of the third conductive layer L3 toward the second end in the width direction Wd. The first end portion 34a, which is the part on the side nearer the first end in the length direction Ld, of the fourth straight part 34 of the fourth conductive layer L4 is connected to this via 35. The fourth conductive layer L4 includes the first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34. The first straight part 31, the second straight part 32, the third straight part 33, and the fourth straight part 34 of the fourth conductive layer L4 are disposed on a plane parallel to the first side surface 25. Similarly to the third conductive layer L3, the fourth conductive layer L4 extends so as to draw a substantially quadrangular path TR formed of the first end portion 34a, the first straight part 31, the third straight part 33, the second straight part 32, and the second end portion 34b. The position of the gap between the first end portion 34a and the second end portion 34b in the fourth conductive layer L4 is nearer the second end in the length direction Ld than the position of the gap between the first end portion 34a and the second end portion 34b in the third conductive layer L3.

(26) The via 35 is connected from an end, which is on the side nearer the first end in the length direction Ld, of the second end portion 34b of the fourth conductive layer L4 toward the second end in the width direction Wd. The first end portion 34a, which is the part on the side nearer the first end in the length direction Ld, of the fourth straight part 34 of the fifth conductive layer L5 is connected to this via 35. The fifth conductive layer L5 includes the first straight part 31, the third straight part 33, and the fourth straight part 34. The first straight part 31, the third straight part 33, and the fourth straight part 34 of the fifth conductive layer L5 are disposed on a plane parallel to the first side surface 25. The fifth conductive layer L5 extends so as to draw a substantially quadrangular path TR formed of the first end portion 34a, the first straight part 31, and the third straight part 33.

(27) In this embodiment, in the inductor wiring 30, each connection part where the first straight part 31 and the third straight part 33 are connected to each other has a substantially chamfered shape when looking in the width direction Wd. Specifically, the surface of the connection part

between the first straight part **31** and the third straight part **33** that is on the side nearer a winding center axis CA of the inductor wiring **30** is a curved surface. Furthermore, the surface of the connection part between the first straight part **31** and the third straight part **33** that is on the opposite side from the winding center axis CA is also a curved surface. Similarly, the surfaces of the connection part between the first straight part **31** and the fourth straight part **34**, the connection part between the second straight part **32** and the third straight part **33**, and the connection part between the second straight part **32** and the fourth straight part **34** that are on the side nearer the winding center axis CA of the inductor wiring **30** are curved surfaces. In addition, the surface of each connection part on the opposite side from the winding center axis CA of the inductor wiring **30** is also a curved surface.

(28) Thus, the inductor wiring **30** is substantially shaped like a wound coil formed of the first to fifth conductor layers L1 to L5 and the vias **35** connecting the first to fifth conductor layers L1 to L5 to each other. The winding center axis CA of the inductor wiring **30** wound in a substantially coil-like shape is aligned with the width direction Wd. In other words, the winding center axis CA is perpendicular to the first side surface **25**. The winding center axis CA is located substantially at the center of the element body **20** when looking in the width direction Wd.

(29) Furthermore, the distance between the first straight parts **31** and the second straight parts **32** is greater than the distance between the third straight parts **33** and the fourth straight parts **34** when looking in the width direction Wd. Therefore, in this embodiment, the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis CA and perpendicular to the first end surface **23** is greater than the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis CA and perpendicular to the mounting surface **21**.

(30) As illustrated in FIG. 4, the first lead-out electrode **40** is connected to a first end, in the extension direction, of the inductor wiring **30**, that is, the end, which is nearer the first end in the length direction Ld, of the third straight part **33** of the first conductive layer L1. As illustrated in FIG. 5, the first lead-out electrode **40** has a substantially quadrangular columnar shape that extends in the height direction Td. The upper end of the first lead-out electrode **40** is flush with the upper surface of the third straight part **33**. As illustrated in FIG. 4, the first lead-out electrode **40** has a substantially square shape when looking in the height direction Td. The dimension of the first lead-out electrode **40** in the length direction Ld is less than or equal to $\frac{1}{4}$ the dimension of the element body **20** in the length direction Ld, and is 48 μm in this embodiment. Furthermore, the dimension of the first lead-out electrode **40** in the width direction Wd is less than or equal to $\frac{1}{4}$ the dimension of the element body **20** in the width direction Wd, and is 48 μm in this embodiment. In this embodiment, the length direction Ld is a direction that is parallel to both the mounting surface **21** and the first side surface **25**.

(31) As illustrated in FIG. 5, the dimension of the first lead-out electrode **40** in the height direction Td is greater than $\frac{1}{2}$ the dimension of the element body **20** in the height direction Td and is 185 μm in this embodiment. The lower surface of the first lead-out electrode **40** in the height direction Td is flush with the mounting surface **21** of the element body **20**. Therefore, the lower surface of the first lead-out electrode **40** in the height direction Td is exposed from the mounting surface **21** of the element body **20**. On the other hand, the upper surface of the first lead-out electrode **40** in the height direction Td is located inside the element body **20**. Since the dimension of the element body **20** in the height direction Td is 200 μm , a distance of 15 μm is secured as the distance from the top surface **22** of the element body **20** to the end, on the side nearer the top surface **22**, of the first lead-out electrode **40** in the height direction Td. In this embodiment, the first lead-out electrode **40** has a substantially quadrangular columnar shape, and therefore, the maximum dimension of the first lead-out electrode **40** in the height direction Td is equal to the dimension of the first lead-out electrode **40** in the height direction Td.

(32) As illustrated in FIG. 4, when looking in the height direction Td, one corner out of the four

corners of the square shape of the first lead-out electrode **40** is aligned with a corner where an imaginary plane containing the first end surface **23** and an imaginary plane containing the first side surface **25** are connected to each other. Among the four side surfaces of the quadrangular columnar shape of the first lead-out electrode **40**, the side surface that is nearer the first end in the length direction L_d is flush with the first end surface **23**. Therefore, the side surface of the first lead-out electrode **40** that is nearer the first end in the length direction L_d is exposed from the first end surface **23** of the element body **20**. In addition, the dimension, in the width direction W_d , of the part of the first lead-out electrode **40** that is exposed from the first end surface **23** is less than or equal to $\frac{1}{4}$ the dimension of the first end surface **23** in the width direction W_d . Therefore, when looking in the length direction L_d , the first lead-out electrode **40** overlaps the inductor wiring **30** only in a region of the first end surface **23** nearer the first side surface **25** than the center of the first end surface **23**.

(33) Among the four side surfaces of the quadrangular columnar shape of the first lead-out electrode **40**, the side surface that is nearer the first end in the width direction W_d is flush with the first side surface **25**. Therefore, the side surface of the first lead-out electrode **40** that is nearer the first end in the width direction W_d is exposed from the first side surface **25** of the element body **20**. In other words, the first lead-out electrode **40** is exposed from the first end surface **23** and the first side surface **25** of the element body **20**. On the other hand, the first lead-out electrode **40** is not exposed from the second end surface **24** and the second side surface **26** of the element body **20**.

(34) In addition, the second lead-out electrode **50** is connected to a second end, in the extension direction, of the inductor wiring **30**, that is, the end, which is nearer the second end in the length direction L_d , of the third straight part **33** of the fifth conductive layer L_5 . The second lead-out electrode **50** has substantially the same shape as the first lead-out electrode **40**. As illustrated in FIG. 5, the second lead-out electrode **50** has a substantially quadrangular columnar shape that extends in the height direction T_d . The upper end of the second lead-out electrode **50** is flush with the upper surface of the third straight part **33**. As illustrated in FIG. 4, the second lead-out electrode **50** has a substantially square shape when looking in the height direction T_d . The dimension of the second lead-out electrode **50** in the length direction L_d is less than or equal to $\frac{1}{4}$ the dimension of the element body **20** in the length direction L_d , and is $48\ \mu\text{m}$ in this embodiment. Furthermore, the dimension of the second lead-out electrode **50** in the width direction W_d is less than or equal to $\frac{1}{4}$ the dimension of the element body **20** in the width direction W_d , and is $48\ \mu\text{m}$ in this embodiment.

(35) As illustrated in FIG. 5, the dimension of the second lead-out electrode **50** in the height direction T_d is greater than $\frac{1}{2}$ the dimension of the element body **20** in the height direction T_d and is $185\ \mu\text{m}$ in this embodiment. The lower surface of the second lead-out electrode **50** in the height direction T_d is flush with the mounting surface **21** of the element body **20**. Therefore, the lower surface of the second lead-out electrode **50** in the height direction T_d is exposed from the mounting surface **21** of the element body **20**. On the other hand, the upper surface of the second lead-out electrode **50** in the height direction T_d is located inside the element body **20**. Since the dimension of the element body **20** in the height direction T_d is $200\ \mu\text{m}$, a distance of $15\ \mu\text{m}$ is secured as the distance from the top surface **22** of the element body **20** to the end, on the side nearer the top surface **22**, of the second lead-out electrode **50** in the height direction T_d . In this embodiment, the second lead-out electrode **50** has a substantially quadrangular columnar shape, and therefore, the maximum dimension of the second lead-out electrode **50** in the height direction T_d is equal to the dimension of the second lead-out electrode **50** in the height direction T_d .

(36) As illustrated in FIG. 4, when looking in the height direction T_d , one corner out of the four corners of the square shape of the second lead-out electrode **50** is aligned with a corner where an imaginary plane containing the second end surface **24** and an imaginary plane containing the second side surface **26** are connected to each other. Among the four side surfaces of the quadrangular columnar shape of the second lead-out electrode **50**, the side surface that is nearer the second end in the length direction L_d is flush with the second end surface **24**. Therefore, the

surface of the second lead-out electrode **50** that is nearer the second end in the length direction **Ld** is exposed from the second end surface **24** of the element body **20**. In addition, the dimension, in the width direction **Wd**, of the part of the second lead-out electrode **50** that is exposed from the second end surface **24** is less than or equal to $\frac{1}{4}$ the dimension of the second end surface **24** in the width direction **Wd**. Therefore, when looking in the length direction **Ld**, the second lead-out electrode **50** overlaps the inductor wiring **30** only in a region of the second end surface **24** nearer the second side surface **26** than the center of the second end surface **24**.

(37) Among the four side surfaces of the quadrangular columnar shape of the second lead-out electrode **50**, the side surface that is nearer the second end in the width direction **Wd** is flush with the second side surface **26**. Therefore, the surface of the second lead-out electrode **50** that is nearer the second end in the width direction **Wd** is exposed from the second side surface **26** of the element body **20**. In other words, the second lead-out electrode **50** is exposed from the second end surface **24** and the second side surface **26**. On the other hand, the second lead-out electrode **50** is not exposed from the first end surface **23** and the first side surface **25**.

(38) Here, in the element body **20**, when looking in the width direction **Wd**, part of the first lead-out electrode **40** overlaps the first straight parts **31** of the inductor wiring **30**. In other words, when looking in the width direction **Wd**, the first straight parts **31** of the inductor wiring **30** are positioned so as to overlap the first lead-out electrode **40**. The extension direction of the first lead-out electrode **40** is parallel to the extension direction of the first straight parts **31**. In addition, among the four side surfaces of the first lead-out electrode **40**, the surface that is on the side nearer the second end in the length direction **Ld** is located on the same plane as the surfaces of the first straight parts **31** that are on the side nearer the second end in the length direction **Ld**. In other words, when looking in the width direction **Wd**, the edge of the first lead-out electrode **40** on the side nearer the winding center axis **CA** of the inductor wiring **30** is aligned with the edges of the first straight parts **31** on the side nearer the winding center axis **CA**. “Aligned” means substantial aligned and, for example, manufacturing errors and so forth of around 10 μm are permitted.

(39) Here, in the element body **20**, when looking in the width direction **Wd**, part of the second lead-out electrode **50** overlaps the second straight parts **32** of the inductor wiring **30**. In other words, when looking in the width direction **Wd**, the second straight parts **32** of the inductor wiring **30** are positioned so as to overlap the second lead-out electrode **50**. The extension direction of the second lead-out electrode **50** is parallel to the extension direction of the second straight parts **32**. In addition, among the four side surfaces of the second lead-out electrode **50**, the surface that is on the side nearer the first end in the length direction **Ld** is located on the same plane as the surfaces of the second straight parts **32** that are on the side nearer the first end in the length direction **Ld**. In other words, when looking in the width direction **Wd**, the edge of the second lead-out electrode **50** on the side nearer the winding center axis **CA** of the inductor wiring **30** is aligned with the edges of the second straight parts **32** on the side nearer the winding center axis **CA**.

(40) As a result of the above-described positional relationship, the distance between the first lead-out electrode **40** and the second lead-out electrode **50** is the same as the distance between the first straight parts **31** and the second straight parts **32** of the inductor wiring **30** when looking in the width direction **Wd**. In other words, when looking in the width direction **Wd**, the distance between the first lead-out electrode **40** and the second lead-out electrode **50** is equal to the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis **CA** of the inductor wiring **30** and perpendicular to the first end surface **23**. “The distance is equal” means that the distance should be substantially equal and, for example, manufacturing errors and so forth of around 10 μm are acceptable.

(41) Out of the outer surfaces of the first lead-out electrode **40**, a first coating layer **70** is stacked on the surfaces that are exposed from the outer surfaces of the element body **20**. In other words, the first coating layer **70** is provided on the mounting surface **21**, the first end surface **23**, and the first side surface **25**. The first coating layer **70** has a two-layer structure consisting of a nickel layer **71**

and a tin layer **72**. The nickel layer **71** composed of nickel is stacked on the surfaces of the first lead-out electrode **40**. The tin layer **72** composed of tin is stacked on the surfaces of the nickel layer **71**.

(42) Out of the outer surfaces of the second lead-out electrode **50**, a second coating layer **80** is stacked on the surfaces that are exposed from the outer surfaces of the element body **20**. In other words, the second coating layer **80** is provided on the mounting surface **21**, the second end surface **24**, and the second side surface **26**. The second coating layer **80** has a two-layer structure consisting of a nickel layer **81** and a tin layer **82**. The nickel layer **81** composed of nickel is stacked on the surfaces of the second lead-out electrode **50**. The tin layer **82** composed of tin is stacked on the surfaces of the nickel layer **81**.

(43) As illustrated in FIG. **3**, the first lead-out electrode **40** and the second lead-out electrode **50** are disposed on a diagonal of the quadrangular shape when the inductor component **10** is viewed in the height direction Td. Therefore, the inductor component **10** has a two-fold symmetrical structure with the center of the mounting surface **21** as the center of rotational symmetry. Therefore, with respect to the position where the first lead-out electrode **40** is exposed from the element body **20**, the second lead-out electrode **50** is exposed from the element body **20** at a two-fold rotationally symmetrical position with the center of the mounting surface **21** being the center of rotation and the height direction Td being the axis of rotation. In addition, as illustrated in FIG. **4**, the internal structure of the element body **20** of the inductor component **10** also has a two-fold symmetrical structure with the center of the element body **20** being the center of rotational symmetry when looking in the height direction Td.

(44) Next, the actions and effects of the above-described embodiment will be described. Effects described below that are common to both the first lead-out electrode **40** and the second lead-out electrode **50** are described using the first lead-out electrode **40** as a representative example.

(45) (1) According to the above-described embodiment, the first lead-out electrode **40** has a substantially columnar shape. In addition, the first lead-out electrode **40** is exposed from the first end surface **23** and the first side surface **25**. In other words, the first lead-out electrode **40** is disposed in the form of a column at a position shifted towards the ridge line between the first end surface **23** and the first side surface **25**. Therefore, the extent to which the inductor wiring **30** disposed inside the element body **20** is covered by the first lead-out electrode **40** can be reduced compared with a hypothetical case where the first lead-out electrode **40** has an L shape that extends onto the mounting surface **21** and covers a large area of the first end surface **23**. As a result, a situation in which magnetic flux generated when a current flows along the inductor wiring **30** is excessively blocked by the first lead-out electrode **40** is suppressed.

(46) (2) According to the above-described embodiment, the maximum dimension of the first lead-out electrode **40** in the height direction Td is greater than $\frac{1}{2}$ the dimension of the element body **20** in the height direction Td. Therefore, the upper end of the first lead-out electrode **40** is located reasonably close to the top surface **22** of the element body **20**. As a result, the first lead-out electrode **40** enables a conductive part to be led out to the mounting surface **21** even when the first end of the inductor wiring **30** in the extension direction is disposed reasonably close to the top surface **22**.

(47) (3) According to the above-described embodiment, the dimension of the first lead-out electrode **40** in the width direction Wd is less than or equal to $\frac{1}{4}$ the dimension of the element body **20** in the width direction Wd. Therefore, when looking in the height direction Td, the region occupied by the first lead-out electrode **40** is reasonably small. As a result, the degree of freedom when designing the wiring path of the inductor wiring **30** inside the element body **20** is increased.

(48) (4) According to the above-described embodiment, the dimension of the first lead-out electrode **40** in the length direction Ld is greater than or equal to $10\ \mu\text{m}$. Therefore, the size of the first lead-out electrode **40** can be guaranteed even if the dicing accuracy is somewhat low when the element body **20** is divided into individual pieces during manufacture of the inductor component

10.

(49) (5) According to the above-described embodiment, the first coating layer **70** is stacked on the surfaces of the first lead-out electrode **40** that are exposed from the element body **20**. Therefore, when mounting the inductor component **10**, it is easy to align the inductor component **10** by using the parts protruding from the surface of the element body **20** as a marker.

(50) (6) According to the above-described embodiment, the first coating layer **70** has a two-layer structure consisting of the nickel layer **71** composed of nickel and the tin layer **72** composed of tin and stacked on the surfaces of the nickel layer **71**. Therefore, damage to the first lead-out electrode **40** caused by melted solder can be prevented by the heat resistance of the nickel layer **71**. In addition, the solder adhesion strength can be increased due to the solder wettability of the tin layer **72** being reasonably high.

(51) (7) According to the above-described embodiment, when looking in the height direction T_d , the inductor component **10** has a two-fold symmetrical structure with the center of the element body **20** being the center of rotational symmetry. In addition, the internal structure of the element body **20** also has a two-fold symmetrical structure with the center of the element body **20** being the center of rotational symmetry when looking in the height direction T_d . Therefore, even if the inductor components **10** are mounted so that the length directions L_d thereof face in opposite directions, the characteristics of the inductor components **10** will be identical. As a result, the orientation of the length direction L_d does not matter when mounting the inductor component **10**.

(52) (8) According to the above-described embodiment, the winding center axis CA of the inductor wiring **30** extends parallel to the width direction W_d . Therefore, when the inductor component **10** is mounted on a circuit board, blocking of magnetic flux at the circuit substrate connected to the mounting surface **21** side of the inductor component **10** can be suppressed compared with a case where the winding center axis CA of the inductor wiring **30** is perpendicular to the mounting surface **21**.

(53) (9) According to the above-described embodiment, the first lead-out electrode **40** is exposed in a region spanning from the first end surface **23** to the first side surface **25**. In other words, the first lead-out electrode **40** is disposed close to the first end in the width direction W_d . Therefore, for example, there is no need to make the paths of the first to fifth conductive layers L_1 to L_5 of the inductor component **10** different from each other in order to secure the distance to the first lead-out electrode **40**, and it is easy to design a path in which the inner diameter of the inductor wiring **30** is the same in each layer.

(54) (10) According to this embodiment, when looking in the width direction W_d , the first lead-out electrode **40** overlaps the first straight parts **31** of the inductor wiring **30**. In other words, the part of the element body **20** that is nearer the second end in the width direction W_d than the first lead-out electrode **40** is effectively utilized as a space in which to place the inductor wiring **30**.

(55) (11) According to the above-described embodiment, when looking in the width direction W_d , the edge of the first lead-out electrode **40** on the side nearer the winding center axis CA of the inductor wiring **30** is aligned with the edges of the first straight parts **31** on the side nearer the winding center axis CA . Therefore, a situation in which magnetic flux passing through the inside of the inductor wiring **30** hits the first lead-out electrode **40** can be avoided.

(56) (12) According to the above-described embodiment, when looking in the width direction W_d , the distance between the first lead-out electrode **40** and the second lead-out electrode **50** is equal to the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis CA of the inductor wiring **30** and perpendicular to the first end surface **23**. In other words, the inner diameter of the inductor wiring **30** is maximized while avoiding a situation in which magnetic flux passing through the inside of the inductor wiring **30** hits the second lead-out electrode **50**.

(57) (13) According to the above-described embodiment, the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis CA and perpendicular to the first

end surface **23** is greater than the inner diameter of the inductor wiring **30** along a straight line passing through the winding center axis CA and perpendicular to the mounting surface **21**. Therefore, in the element body **20**, which is longer in the length direction Ld than in the height direction Td, when looking in the width direction Wd, the length of wiring path of the inductor wiring **30** can be increased as a result of the diameter of the wiring path of the inductor wiring **30** being increased.

(58) (14) According to the above-described embodiment, each layer of the second to fourth conductive layers L2 to L4 of the inductor wiring **30** includes a first straight part, a second straight part, a third straight part, and a fourth straight part, which are disposed on planes that are parallel to the first side surface **25** inside the element body **20**. Therefore, the path length of the inductor wiring **30** per unit volume of the element body **20** can be increased.

(59) (15) According to the above-described embodiment, in the inductor wiring **30**, the surfaces of the connection part between the first straight part **31** and the third straight part **33**, the connection part between the first straight part **31** and the fourth straight part **34**, the connection part between the second straight part **32** and the third straight part **33**, and the connection part between second straight part **32** and the fourth straight part **34** that are on the side near the winding center axis CA of the inductor wiring **30** are curved surfaces. Therefore, when the current flowing along the inductor wiring **30** changes direction by 90 degrees, current loss can be suppressed due to the direction changing gradually.

(60) (16) According to the above-described embodiment, therefore, when looking in the length direction Ld, the first lead-out electrode **40** overlaps the inductor wiring **30** only in a region of the first end surface **23** that is nearer the first side surface **25** than the center of the first end surface **23**. In this case, the first lead-out electrode **40** is not disposed at a position where the inductor wiring **30** does not overlap the first lead-out electrode **40** when looking in the length direction Ld. Therefore, the first lead-out electrode **40** is less likely to block magnetic flux generated when a current flows along the inductor wiring **30**.

(61) The above-described embodiment can be modified in the following ways. The embodiment and the following modifications can be combined with each other to the extent that they are not technically inconsistent.

(62) The size of the element body **20** is not limited to the example given in the embodiment. For example, the dimensions of the element body **20** in the respective directions may be a dimension of 600 μm in the length direction Ld, a dimension of 300 μm in the width direction Wd, and a dimension of 300 μm in the height direction Td, or may be a dimension of 250 μm in the length direction Ld, a dimension of 125 μm in the width direction Wd, and a dimension of 125 μm in the height direction Td. In addition, for example, the dimension in the height direction Td and the dimension in the width direction Wd do not have to be equal to each other and the dimension in the height direction Td may be larger than the dimension in the length direction Ld.

(63) The shape of the inductor wiring **30** does not have to be a quadrangular columnar shape. The shape of the inductor wiring **30** may be a polygonal columnar shape other than a quadrangular columnar shape or may be a cylindrical shape.

(64) It is not essential that there be a distance from the lower ends of the fourth straight parts **34** of the inductor wiring **30** to the mounting surface **21**, but this distance is preferably from around 10 μm to around 20 μm . The path TR of the inductor wiring **30** can be made larger, the more the distance from the lower ends of the fourth straight parts **34** to the mounting surface **21** is decreased. On the other hand, when the element body **20** is divided into individual pieces when manufacturing the inductor component **10**, the risk of the inductor wiring **30** being exposed from the outer surface of the element body **20** is suppressed even when the dicing precision is quite low by making the distance from the lower end of the fourth straight part **34** to the mounting surface **21** reasonably large. In addition, similarly, it is also not essential that there be a distance from the upper ends of the third straight parts **33** of the inductor wiring **30** to the top surface **22**, but this distance is

preferable from around 10 μm to around 20 μm .

(65) In each connection part between the first straight part **31** and the third straight part **33**, the surface on the side nearer the winding center axis CA of the inductor wiring **30** and the surface on the opposite side from the winding center axis CA do not have to be curved surfaces. For example, the surfaces may have an angle that bends through 90 degrees, or the first straight part **31** and the third straight part **33** may be connected by an inclined part that is inclined to both the first straight part **31** and the third straight part **33**. In this case, it is easier to increase the length of the inductor wiring **30** by making the straight parts of the first and third straight parts **31** and **33** longer. This point similarly applies to the connection parts between the other straight parts.

(66) Regarding the inductor wiring **30**, it is not necessary that all of the first to fourth straight parts **34** be disposed on a plane parallel to the first side surface **25** inside the element body **20**. For example, only the third straight part **33** and the second straight part **32** may be disposed in the first conductive layer L1 and only the fourth straight part **34** and the first straight part **31** may be disposed in the second conductive layer L2. In this way, even if the inductor wiring **30** is formed so that only some of the straight parts are disposed in a single conductive layer, the inductor wiring **30** may be wound as a whole.

(67) In the inductor wiring **30**, the distance between the third straight part **33** and the fourth straight part **34** may be greater than or equal to the distance between the first straight part **31** and the second straight part **32** when looking in the width direction Wd. The distances between the straight parts may be changed as appropriate in accordance with the shape of the element body **20** and the required electrical characteristics.

(68) The positional relationship between the first straight parts **31** and the first lead-out electrode **40** is not limited to the example given in the embodiment. The surfaces of the first straight parts **31** on the side nearer the winding center axis CA do not have to be disposed on the same plane as the surface of the first lead-out electrode **40** on the side nearer the second end in the length direction Ld. In addition, for example, the first straight parts **31** do not have to overlap the first lead-out electrode **40** when looking in the width direction Wd. This point similarly applies to the positional relationship between the second straight parts **32** and the second lead-out electrode **50**.

(69) The winding center axis CA of the inductor wiring **30** does not have to extend in a direction perpendicular to the first side surface **25**. For example, the winding center axis CA may extend in a direction perpendicular to the first end surface **23** or may extend in a direction perpendicular to the mounting surface **21**. When the winding center axis CA extends in a direction perpendicular to any of the outer surfaces of the element body **20**, it is easy to wind the inductor wiring **30** having a uniform diameter inside the element body **20**.

(70) The shape of the inductor wiring **30** when looking in the width direction Wd, i.e., the path TR of the inductor wiring **30** is not limited to the example given in the embodiment. For example, the path TR of the inductor wiring **30** may have a polygonal shape other than a quadrangular shape or may have an elliptical or circular shape when looking in the width direction Wd.

(71) The shape of the inductor wiring **30** is not limited to the example given in the embodiment. For example, the inductor wiring **30** does not have to be shaped like a coil and may instead have a straight line shape or a meandering shape.

(72) The dimensions of the first lead-out electrode **40** are not limited to the examples given in the embodiment. The smaller the dimension of the first lead-out electrode **40** in each direction, the smaller the amount of magnetic flux that will be blocked by the first lead-out electrode **40**. On the other hand, for example, it is preferable that the dimensions of the first lead-out electrode **40** in the width direction Wd and the length direction Ld be around 10 μm or higher in order to dice the element body **20**. In addition, in the example illustrated in FIG. 8, in an inductor component **110**, the dimension of a first lead-out electrode **140** in the length direction Ld is greater than $\frac{1}{4}$ the dimension of the element body **20** in the length direction Ld. In particular, in this example, the dimension of the first lead-out electrode **140** in the length direction Ld is greater than the distance

from the first end surface of the element body **20** in the length direction Ld to the surfaces of the first straight parts **31** on the side nearer the winding center axis CA. Similarly, the dimension of a second lead-out electrode **150** in the length direction Ld is greater than $\frac{1}{4}$ of the dimension of the element body **20** in the length direction Ld. In this example, the edge of the first lead-out electrode **140** on the side near the winding center axis CA is located nearer the winding center axis CA than the edges of the first straight parts **31** on the side near the winding center axis CA. In addition, the edge of the second lead-out electrode **150** on the side near the winding center axis CA is located nearer the winding center axis CA than the edges of the second straight parts **32** on the side near the winding center axis CA. In this modification, the areas over which the first lead-out electrode **140** and the second lead-out electrode **150** are exposed from the mounting surface **21** can be increased, and therefore, the inductor component **110** is more easily mounted on a substrate.

(73) Regarding the position of the first lead-out electrode **40**, when looking in the height direction Td, one corner out of the four corners of the square shape of the first lead-out electrode **40** does not have to be aligned with a corner where an imaginary plane containing the first end surface **23** and an imaginary plane containing the first side surface **25** are connected to each other. The first lead-out electrode **40** may be exposed from at least the mounting surface **21**, the first end surface **23**, and the first side surface **25**. For example, part of the first lead-out electrode **40** may be disposed nearer the first end in the length direction Ld than an imaginary plane containing the first end surface **23** and may be disposed nearer the first end in the width direction Wd than an imaginary plane containing the first side surface **25**. This point similarly applies to the second lead-out electrode **50**.

(74) The dimensions of the first lead-out electrode **40** are not limited to the examples given in the embodiment. For example, the dimension of the first lead-out electrode **40** in the height direction Td may be less than or equal to $\frac{1}{2}$ the dimension of the element body **20** in the height direction Td. Furthermore, the dimension of the first lead-out electrode **40** in the height direction Td may be equal to the dimension of the element body **20** in the height direction Td. In this case, the first lead-out electrode **40** is exposed at the top surface **22** of the element body **20** as well.

(75) So long as the first lead-out electrode **40** has a columnar shape, the first lead-out electrode **40** may locally include parts that have a larger dimension in the height direction Td. In this case, it is suitable that the maximum dimension of the first lead-out electrode **40** in the height direction Td be less than or equal to $\frac{1}{2}$ the dimension of the element body **20** in the height direction Td. The maximum dimension may be measured by performed electron microscopy on a cross-section perpendicular to the mounting surface **21** including the part where the dimension is larger in the height direction Td.

(76) The positional relationship between the first lead-out electrode **40** and the inductor wiring **30** is not limited to the example in the embodiment. For example, the first lead-out electrode **40** does not have to overlap the inductor wiring **30** when looking in the length direction Ld. In this case, the element body **20** can be disposed at a location that the inductor wiring **30** does not overlap the first lead-out electrode **40** when looking in the length direction Ld. As a result, the volume of the element body **20** of the inductor component **10** can be increased.

(77) The shape of the second lead-out electrode **50** does not have to be the same as that of the first lead-out electrode **40**. In addition, the positional relationship between the second lead-out electrode **50** and the first lead-out electrode **40** is not limited to the example given in the embodiment. In other words, the second lead-out electrode **50** and the first lead-out electrode **40** do not have to be in a positional relationship having two-fold symmetry with the center of the element body **20** being the center of rotational symmetry when looking in the height direction Td.

(78) The element body **20** does not have to be completely formed of the same material, and for example, only the surface layers of the first side surface **25** and the second side surface **26** may be colored. In this case, it is easy to see whether or not the mounting surface **21** of the inductor component **10** is facing toward the circuit board when mounting the inductor component **10**.

(79) The structure of the first coating layer **70** is not limited to the example given in the

embodiment. For example, the first coating layer **70** may be formed by plating, by applying a metal paste and sintering the metal paste, or may consist only of the tin layer **72**. In addition, the first coating layer **70** may be omitted. In the case where the first coating layer **70** is omitted, the part of the first lead-out electrode **40** that is exposed from the element body **20** functions as a terminal part for a substrate and so forth. This point similarly applies to the second coating layer **80**.

(80) While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. An inductor component comprising: an element body comprising an insulator, the element body having a mounting surface and a top surface that are parallel to each other, a first end surface and a second end surface that are parallel to each other, and a first side surface and a second side surface that are parallel to each other; an inductor wiring disposed inside the element body, the inductor wiring being shaped like a coil wound inside the element body, and a center axis of winding of the inductor wiring being perpendicular to the first side surface; a first lead-out electrode connected to a first end of the inductor wiring, the first lead-out electrode having a columnar shape that extends in a direction perpendicular to the mounting surface and being exposed from the first end surface and the first side surface, an upper end of the first lead-out electrode being flush with an upper surface of a straight part of the inductor wiring; and a second lead-out electrode connected to a second end of the inductor wiring, wherein part of the first lead-out electrode and part of the second lead-out electrode are exposed from the mounting surface, a portion of the first side surface abuts and extends along the first lead-out electrode from an upper surface of the first lead-out electrode to the mounting surface of the element body, a portion of the first end surface abuts and extends along the first lead-out electrode from the upper surface of the first lead-out electrode to the mounting surface of the element body, and the first lead-out electrode is not exposed from the second end surface and the second side surface.
2. The inductor component according to claim 1, wherein a maximum dimension of the first lead-out electrode in a direction perpendicular to the mounting surface is greater than $\frac{1}{2}$ a dimension of the first end surface in a direction perpendicular to the mounting surface.
3. The inductor component according to claim 1, wherein the first lead-out electrode has a quadrangular columnar shape, and a dimension, in a direction perpendicular to the first side surface, of a part of the first lead-out electrode exposed from the first end surface is less than or equal to $\frac{1}{4}$ a dimension of the first end surface in a direction perpendicular to the first side surface.
4. The inductor component according to claim 1, wherein the first lead-out electrode has a quadrangular columnar shape, and a dimension of the first lead-out electrode in a direction perpendicular to the first end surface is greater than or equal to 10 μm .
5. The inductor component according to claim 1, wherein a first coating layer is stacked on at least a surface of the first lead-out electrode that is exposed from the mounting surface.
6. The inductor component according to claim 5, wherein the first coating layer includes a nickel layer stacked on a surface of the first lead-out electrode that is exposed from the element body, and a tin layer stacked on a surface of the nickel layer.
7. The inductor component according to claim 1, wherein the second lead-out electrode has a columnar shape that extends in a direction perpendicular to the mounting surface, and is exposed from the second end surface and the second side surface.
8. The inductor component according to claim 7, wherein a shape of the second lead-out electrode is identical to a shape of the first lead-out electrode, and the second lead-out electrode is exposed from the element body at a two-fold rotationally symmetrical position, with respect to a position

where the first lead-out electrode is exposed from the element body, with a center of the mounting surface being a center of rotation and a direction perpendicular to the mounting surface being an axis of rotation.

9. The inductor component according to claim 1, wherein the first lead-out electrode does not overlap the inductor wiring when looking in a direction perpendicular to the first end surface.

10. The inductor component according to claim 1, wherein the first lead-out electrode overlaps the inductor wiring only in a region of the first end surface nearer the first side surface than a center of the first end surface.

11. An inductor component comprising: an element body comprising an insulator, the element body having a mounting surface and a top surface that are parallel to each other, a first end surface and a second end surface that are parallel to each other, and a first side surface and a second side surface that are parallel to each other; an inductor wiring disposed inside the element body; a first lead-out electrode connected to a first end of the inductor wiring, the first lead-out electrode having a columnar shape that extends in a direction perpendicular to the mounting surface and being exposed from the first end surface and the first side surface; and a second lead-out electrode connected to a second end of the inductor wiring, wherein part of the first lead-out electrode and part of the second lead-out electrode are exposed from the mounting surface, a portion of the first side surface abuts and extends along the first lead-out electrode from an upper surface of the first lead-out electrode to the mounting surface of the element body, a portion of the first end surface abuts and extends along the first lead-out electrode from the upper surface of the first lead-out electrode to the mounting surface of the element body, the first lead-out electrode is not exposed from the second end surface and the second side surface, the inductor wiring is shaped like a coil wound inside the element body, a center axis of winding of the inductor wiring is perpendicular to the first side surface, and the first lead-out electrode and part of the inductor wiring overlap when looking in a direction in which the center axis extends.

12. The inductor component according to claim 11, wherein an extension direction of the first lead-out electrode and an extension direction of the inductor wiring are parallel to each other at an overlapping location, and an edge, which is on a side near the center axis, of the first lead-out electrode is located nearer the center axis than an edge, on a side near the center axis, of the part of the inductor wiring at the overlapping location when looking in the direction in which the center axis extends.

13. The inductor component according to claim 11, wherein in a direction perpendicular to the mounting surface, a distance from the top surface to an end, which is on a side near the top surface, of the first lead-out electrode, a distance from the top surface to an end, which is on a side near the top surface, of the inductor wiring, and a distance from the mounting surface to an end, which is on a side near the mounting surface, of the inductor wiring are all from 10 μm to 20 μm .

14. An inductor component comprising: an element body having a mounting surface and a top surface that are parallel to each other, a first end surface and a second end surface that are parallel to each other, and a first side surface and a second side surface that are parallel to each other; an inductor wiring disposed inside the element body; a first lead-out electrode connected to a first end of the inductor wiring, the first lead-out electrode having a columnar shape that extends in a direction perpendicular to the mounting surface and being exposed from the first end surface and the first side surface; and a second lead-out electrode connected to a second end of the inductor wiring, wherein part of the first lead-out electrode and part of the second lead-out electrode are exposed from the mounting surface, the inductor wiring is shaped like a coil wound inside the element body, a center axis of winding of the inductor wiring is perpendicular to the first side surface, the first lead-out electrode and part of the inductor wiring overlap when looking in a direction in which the center axis extends, an extension direction of the first lead-out electrode and an extension direction of the inductor wiring are parallel to each other at an overlapping location where the first lead-out electrode and the part of the inductor wiring overlap when looking in the

direction in which the center axis extends, and an edge, which is on a side near the center axis, of the first lead-out electrode and an edge, which is on a side near the center axis, of the inductor wiring are aligned at the overlapping location when looking in the direction in which the center axis extends.

15. The inductor component according to claim 14, wherein the second lead-out electrode has a columnar shape that extends in a direction perpendicular to the mounting surface, and is exposed from the second end surface and the second side surface, and when looking in the direction in which the center axis extends, a distance between the first lead-out electrode and the second lead-out electrode is equal to an inner diameter of the inductor wiring along a straight line passing through the center axis and perpendicular to the first end surface.

16. The inductor component according to claim 15, wherein the inductor wiring includes a first straight part that is disposed nearer the first end surface than a center of the element body and extends in a direction perpendicular to the mounting surface, and a second straight part that is disposed nearer the second end surface than the center of the element body and extends in a direction perpendicular to the mounting surface, and an inner diameter of the inductor wiring along a straight line that passes through the center axis and is perpendicular to the first end surface is greater than an inner diameter of the inductor wiring along a straight line that passes through the center axis and is perpendicular to the mounting surface, and is equal to a distance between the first straight part and the second straight part.

17. The inductor component according to claim 16, wherein the inductor wiring includes a third straight part that is disposed nearer a side opposite the mounting surface than the center of the element body and that extends in a direction perpendicular to the first end surface, and a fourth straight part that is disposed nearer the mounting surface than the center of the element body and that extends in a direction perpendicular to the first end surface, and the first straight part, the second straight part, the third straight part, and the fourth straight part are disposed on a plane parallel to the first side surface inside the element body.

18. The inductor component according to claim 17, wherein surfaces of a connection part between the first straight part and the third straight part, a connection part between the first straight part and the fourth straight part, a connection part between the second straight part and the third straight part, and a connection part between the second straight part and the fourth straight part that are on a side near the center axis are curved surfaces.
